Vekua boundary data in triply connected domains

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1 Statement and argument

The argument below settles the case of three boundary components in the stated analytic-boundary problem. The full catalogue target, which includes every connectivity $n \geq 3$, remains unresolved here for $n \geq 4$.

Let Ω be a bounded connected planar domain bounded by n pairwise disjoint real-analytic Jordan curves. Label the outer curve Γ_0 and the inner curves $\Gamma_1, \dots, \Gamma_{n-1}$. Suppose a nonconstant real harmonic function $u \in C^2(\Omega) \cap C^1(\overline{\Omega})$ satisfies

$$u = c_j, \qquad \partial_\nu u = d_j \qquad \text{on } \Gamma_j, \tag{1}$$

where the c_j, d_j are real constants and ν is the outward unit normal of Ω. Every d_j is nonzero: constant Dirichlet data and zero normal derivative on one analytic boundary arc would make u constant by harmonic reflection and Cauchy uniqueness.

Lemma 1.1 (A holomorphic primitive with bounded valence)**.** *Put $q = u_x - iu_y$. Then q has exactly $n-2$ zeros in Ω, counted with multiplicity. There is a single-valued holomorphic function F on Ω, continuous up to its boundary, such that*

$$F' = q^2, \qquad F(z) = -d_j^2 \overline{z} + C_j \quad (z \in \Gamma_j) \tag{2}$$

for constants $C_j \in \mathbb{C}$. For every $w \notin F(\partial\Omega)$, the number of solutions of $F(z) = w$ in Ω, counted with multiplicity, is at most $n-1$.

Proof. Orient every boundary component positively with respect to Ω: the outer component is counterclockwise and the inner components are clockwise. Let $\tau = dz/ds$ be the complex unit tangent in this orientation. Then $\nu = -i\tau$. Because u is constant along a boundary component, its tangential derivative is zero and its gradient is $d_j\nu$. Consequently

$$q = d_j\overline{\nu} = id_j\overline{\tau} \qquad \text{on } \Gamma_j. \tag{3}$$

The winding of q about zero is -1 on the outer component and $+1$ on each inner component. The turning-number theorem for smooth Jordan curves and the argument principle therefore give

$$\#\{q = 0\} \text{ with multiplicity} = -1 + (n-1) = n - 2. \tag{4}$$

These uses of the argument principle are legitimate up to the boundary: locally straighten an analytic boundary arc conformally; $u - c_j$ has zero boundary values and extends harmonically by

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odd reflection. Thus q extends holomorphically across each boundary arc and, by $d_j \neq 0$, has no boundary zero.

Squaring (3) and using $dz = \tau ds$ gives

$$q^2\,dz = -d_j^2\overline{\tau}\,ds = -d_j^2\,d\overline{z} \qquad \text{on } \Gamma_j. \tag{5}$$

In particular every boundary period of $q^2\,dz$ is zero. The inner boundary cycles generate the first homology of Ω, so all its periods vanish. Integrating $q^2\,dz$ defines a single-valued holomorphic primitive F; integrating (5) along each component proves (2).

Let D_0 be the bounded Jordan domain enclosed by Γ_0, and let D_j be the hole enclosed by Γ_j for $j \geq 1$. Define the reflected and scaled Jordan domains

$$K_j = \{-d_j^2\overline{z} + C_j : z \in D_j\}.$$

They need not be disjoint. The map in this display reverses orientation. Hence $F(\Gamma_0)$ winds clockwise around K_0, and each $F(\Gamma_j)$, $j \geq 1$, winds counterclockwise around K_j. For w off all these boundary images, the argument principle gives the exact formula

$$N_F(w) = \sum_{j=1}^{n-1} \mathbf{1}_{K_j}(w) - \mathbf{1}_{K_0}(w). \tag{6}$$

The left side counts zeros of $F - w$ with multiplicity, since F has no poles. In particular it is nonnegative and is at most $n - 1$. □

Theorem 1.2 (Triply connected nonexistence)**.** *No nonconstant harmonic function can satisfy* (1) *on a bounded triply connected domain with real-analytic Jordan boundary components.*

Proof. For $n = 3$, (4) says that q has one zero a, of multiplicity one. Since $F' = q^2$, the holomorphic map F has local degree three at a. More explicitly,

$$F(z) - F(a) = c(z-a)^3 + O((z-a)^4), \qquad c \neq 0.$$

Choose a sufficiently small closed disc about a contained in Ω. For every w sufficiently close to $F(a)$, Rouché's theorem gives three solutions of $F(z) = w$ in this disc, counted with multiplicity. Choose such a w outside the finite union $F(\partial\Omega)$ of Jordan curves; this is possible because that union has empty interior. Then $N_F(w) \geq 3$, contradicting $N_F(w) \leq 2$ from (6). □

Corollary 1.3 (A restriction at higher connectivity)**.** *Under* (1)*, if q has a zero of multiplicity $k \geq 1$, then*

$$2k + 1 \leq n - 1. \tag{7}$$

In particular, a nonconstant solution at connectivity $n \geq 3$ would have at least two distinct interior critical points. At connectivity four or five all zeros of q would be simple.

Proof. The local degree of F at such a zero is $2k + 1$. The same nearby value argument bounds it by the global valence $n - 1$. Since the sum of the multiplicities is $n - 2$, (7) excludes a single distinct critical point and gives the last assertions. □

The underlying harmonic problem and the annular classification appear in [1, Section 5, Theorem 5.5]. No sign restriction on the separate fluxes was imposed above: their squares enter (2), while their nonzero signs change the argument of q only by a constant on each boundary component.

References

[1] D. Khavinson, A. Yu. Solynin and D. Vassilev, *Overdetermined boundary value problems, quadrature domains and applications*, Comput. Methods Funct. Theory **5** (2005), 19–48, §5. https://www.unm.edu/~vassilev/quadr.pdf.